Learning Objectives

This lab lets you practice writing programs using dictionaries.

Estimated Size

1 program file: `loops_and_dictionaries.py`, 2 functions, and less than 15 lines of code in each function.

Setup

In your Python files, you must include author information. On the first line, add a `# Author` comment line, where you mention your full name. Similarly, include your email and Spire ID on the `# Email` and `# Spire ID` comment lines, respectively. Remember, these comments must appear **at the beginning of every Python file** that you submit to Gradescope. For example:

```
# Author   : John Snow
# Email    : jsnow@umass.edu
# Spire ID : 12345678
```

1. Implement function `pyramid` (8 points)

Write a function called **`pyramid(n)`** that generates the following pattern:

Given a positive integer n, print n lines.

- The first line should have numbers from n down to 1. Then have a newline **`('\n')`**. The newline is a special character that can be used to print a new line when used inside a string. For example, try printing the string **`'abc\ndef'`**
- The second line should start from n-1 down to 1. Then have a newline **`('\n')`**
- Continue until the last line has just 1. Then have a newline **`('\n')`**
- Each line's numbers should be separated by a **single space**.

Below are examples of running a correct implementation of this function:

```
print(pyramid(4))

4 3 2 1
3 2 1
2 1
1
```

```
print(pyramid(5))

5 4 3 2 1
4 3 2 1
```

```
3 2 1
2 1
1
```

2. Implement function `merge_dicts` (7 points)

We often need to combine two dictionaries that have overlapping keys. Write a function, called **`merge_dicts`**, which should:

- Take two dictionaries, `d1` and `d2`, as parameters.
- Return a **new dictionary** that contains all keys from both `d1` and `d2`.
- If a key appears in both dictionaries, its value in the result should be the **sum** of the two values.

Below are examples of running a correct implementation of this function:

```
print(merge_dicts({'a': 1, 'b': 2}, {'b': 3, 'c': 4}))
# {'a': 1, 'b': 5, 'c': 4}

print(merge_dicts({'x': 10}, {'y': 20}))
# {'x': 10, 'y': 20}

print(merge_dicts({}, {'a': 5}))
# {'a': 5}

print(merge_dicts({'a': 1, 'b': 2, 'c': 3}, {'b': 5, 'd': 10}))
# {'a': 1, 'b': 7, 'c': 3, 'd': 10}
```

Hints

- You can start with a **copy** of `d1` using `.copy()` and then update it using a loop.
- Check if a key exists in both dictionaries using the `in` keyword.

4. Submission `to` Gradescope

Log in to [Gradescope](), select **Lab 07: (Code)**, and submit the required file. Wait until the autograder finishes running and returns the result to you. You can resubmit as many times as you like before the deadline.

(Common Paragraph) Academic Honesty

All work that is completed in this assignment is your own (if you work individually) or your own group's (if you work in a group). You may talk to other students about the problems, and brainstorm about solutions. However, you may NOT share code in any way, except with your group members. What you submit **must be your own or your own group's work**.

You may not use any code that is posted on the internet. If you are not sure it is in your best interest to contact the course staff. We will be using software that will compare your code to other students in the course as well as online resources. It is very easy for us to detect similar submissions and will result in a failure for the exercise or possibly a failure for the course. Please, do not do this. It is important to be academically honest and submit your work only. Please review the UMass Academic Honesty Policy and Procedures so you are aware of what this means.

Copying solutions, whether partial or whole, obtained from other students or elsewhere, is academic dishonesty. Do not share your code with your classmates, and do not use your classmates' code. If you are confused about what constitutes academic dishonesty you should re-read the course policies. We assume you have read the course policies in detail and by submitting this project you have provided your virtual signature in agreement with these policies.